

GAWTAM CHITHRA RAMESH

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Education

KTH Royal Institute of Technology

MSc Systems, Controls & Robotics – GPA 4.62/5.0

Aug 2024 – Jun 2026

Stockholm, Sweden

Indian Institute of Technology Madras

Dual Degree B.Tech Engineering Design + M.Tech Robotics – GPA 7.9/10

Aug 2019 – Jun 2024

Chennai, India

- Awarded (among 30 of 800+) the IIT Madras **Young Research Fellowship** based on research potential.
- Secured **97.64** percentile JEE Advanced; **All India Rank 52** in UCEED among 50,000+ candidates.

Skills: Python, C++, PyTorch, JAX, OpenCV, YOLO, ROS/ROS2, MuJoCo, PyBullet, MATLAB, Docker, Git

Professional Experience

Device Research Intern – 3D Scene Graphs

Jul 2025 – Dec 2025

Ericsson Research

Lund, Sweden

- Integrated instance segmentation + tracking into Hydra (SOTA 3DSG framework) for robust scene interpretation.
- Built RGB-D inpainting pipeline to remove privacy-sensitive objects from 3D scene graphs in real time.
- Benchmarked SOTA inpainting methods on reconstruction fidelity vs. real-time performance trade-off.
- **Lead author:** IMPROVE 2026 (Long Oral, Best Paper nominee); ICPR 2026 (under review).

Multi-Object Detection & Tracking Intern

Jun 2021 – Jul 2021

Blurgs Research Labs

Remote, India

- Analyzed 5 SOTA MOT networks for an Indian Navy drone/CCTV surveillance product.
- Trained Siam-MOT and FairMOT; deployed FairMOT for real-time multi-object tracking and re-ID.

Research Experience

Taxonomy-Guided VLMs for Deformable Object Manipulation

Jan 2025 – Nov 2025

KTH Robotics, Perception & Learning Lab – Advisor: Prof. Florian T. Pokorny

Stockholm, Sweden

- Integrated VLMs (GPT-4o, Gemini, Qwen) with T-DOM taxonomy to map scenes to robot motion primitives.
- Built prompt-taxonomy framework; evaluated with F1-Score, BERT, BLEU/ROUGE on taxonomy correctness.
- **Lead author:** ROMADO @ IROS 2025 (accepted); targeting IROS 2026 + IJRR full paper.

Autonomous Ultrasound Scan via Structure-from-Motion

Jan 2024 – May 2024

IIT Madras INSPIRE Lab – Advisor: Prof. Nirav Patel

Chennai, India

- COLMAP SfM (monocular RGB, 20 poses) + OpenMVS dense reconstruction of human torso; no calibration needed.
- Point cloud segmentation (abdominal ROI) + surface normal estimation; ROS + MoveIt motion planning.

Technical Projects

3D Open-Vocabulary Semantic Mapping | DD2600 Robot Learning, KTH

2025

- Object presence/absence detection in CLIP voxel maps (ARKitScenes; OwlV2 + SAM + CLIP pipeline).
- ResNet18 classifier: 86.2% val accuracy, ROC-AUC 0.917; voxel-threshold baseline: 85%, AUC 0.921.

Image Analysis & Computer Vision | DD2423, KTH

2024

- Gaussian FFT filtering; differential-geometry edge detection (L_{vv}/L_{vvv} zero-crossings); Hough transform.
- SIFT + RANSAC for homography + fundamental matrix estimation; 3D reconstruction via triangulation.

Deep Learning Advanced | DD2610, KTH

2025

- SimCLR: NTXent loss, ResNet18 + projection head, 86% linear-eval accuracy on 1% CIFAR-10 labels.
- MAE: ViT from scratch (PatchEmbed, multi-head attention, asymmetric encoder-decoder, 75% masking).

Lead Author Publications

- Robust Object-layer Construction of 3D Scene Graphs Using Instance Segmentation. **IMPROVE 2026 (Best Paper nominee)**.
- Synchronized RGB-D Inpainting for Privacy-Aware 3D Scene Graph Construction. **ICPR 2026 (Under Review)**.
- Scene Understanding in Deformable Object Manipulation via Taxonomy-Guided VLMs. **IROS 2025**.